

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.3: Cell Biology — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 1.3: Cell Biology
Driving Question	DRIVING QUESTION: "Why do the cells in my body — and in the sukuma wiki on my plate — look so different from each other, and how does each cell's structure allow it to do its specific job?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is a Cell? Launching the Phenomenon — Blood, Sukuma Wiki, and the Mystery of Life's Building Blocks	Students examined prepared microscope slides (or projected micrographs) of human blood cells and sukuma wiki leaf cells side by side. They observed that blood cells appear small, round, and reddish with no visible green structures, while sukuma wiki cells appear larger, box-like, and packed with green chloroplasts. Teacher should prompt students to record specific visual differences (shape, colour, internal structures) before any instruction is given — this activates prior knowledge and surfaces misconceptions.	Students learned that cells can differ dramatically in shape, size, and internal structures while still being classified as cells. They were introduced to the cell as the basic structural and functional unit of all living organisms. Key vocabulary established: cell, organelle, cell membrane, cell wall, nucleus, chloroplast. Students also learned that both human body cells and plant cells (sukuma wiki) are eukaryotic — they contain a membrane-bound nucleus — even though they look very different.	Students explained that the visible differences between blood cells and sukuma wiki cells are not random — each structural feature serves a function tied to what that cell must do. Blood cells are shaped to carry oxygen efficiently; sukuma wiki cells contain chloroplasts because they must capture sunlight for photosynthesis. This lesson anchors the Driving Question: structure always serves function, and that principle will be revisited and deepened in every subsequent lesson. Teacher note: record all student questions on the Driving Question Board (DQB) — unanswered questions here will be resolved across Lessons 2–8.
Lesson 2 The Cell Membrane and Cell Wall — What Controls What Gets In and Out?	Students observed a demonstration or simulation of osmosis using raw sukuma wiki leaves placed in plain water versus concentrated salt water (or a simple model using dialysis tubing). They observed that the leaf in salt water wilts and loses rigidity, while the leaf in plain water remains turgid and firm. Teacher should ensure students record what they see before explaining why	Students learned that every cell is bounded by a cell membrane (phospholipid bilayer) that is selectively permeable — it controls which substances enter and leave the cell. Plant cells (like sukuma wiki) have an additional rigid cell wall made of cellulose, which provides structural support and prevents over-expansion. Key vocabulary: selective permeability, phospholipid bilayer,	Students explained the wilting observation using the concept of osmosis and turgor pressure: when salt water draws water out of sukuma wiki cells, turgor pressure drops and the cell wall can no longer be pushed outward, causing wilting. They connected this to the Driving Question by noting that the presence of a cell wall in plant cells (but not animal cells) is a structural difference

	— direct observation precedes explanation. A diagram of the fluid mosaic model of the cell membrane may also be shown via projected image.	cell wall, cellulose, turgor pressure, osmosis, diffusion. Students learned that the cell wall is why sukuma wiki cells maintain their box-like shape under the microscope.	that directly explains the different shapes observed in Lesson 1. Teacher note: cold-call at least 3 students to articulate why animal cells do NOT have a cell wall and what consequence that has for their shape.
Lesson 3 What Is Inside the Cell Doing All the Work? — Organelle Structure & Function (Animal Cell)	Students examined a detailed labelled diagram and/or 3D model of an animal cell, identifying 10 key organelles: nucleus, nucleolus, mitochondria, endoplasmic reticulum (rough and smooth), Golgi apparatus, ribosomes, lysosomes, vacuoles, cytoplasm, and cell membrane. A 'Nairobi Industrial Park' analogy was used — each organelle was mapped to a department in a factory (e.g., nucleus = head office, mitochondria = power generator, Golgi apparatus = packaging and dispatch). Students observed and recorded each organelle's shape and location within the cell.	Learners identified and explained the structure and function of 10 animal cell organelles, mapping each to a department in a Nairobi Industrial Park factory. Key functions: nucleus stores genetic information and controls cell activity; mitochondria produce ATP energy via cellular respiration; ribosomes synthesise proteins; rough ER transports proteins; smooth ER synthesises lipids; Golgi apparatus packages and ships proteins; lysosomes digest waste; vacuoles store substances; cytoplasm suspends organelles; cell membrane controls entry and exit. Teacher note: students often confuse the nucleus with the nucleolus — clarify that the nucleolus (inside the nucleus) produces ribosomal RNA.	Students explained that the internal complexity of an animal cell — all those specialised organelles — is what allows a single cell to carry out dozens of simultaneous tasks. They connected this to the Driving Question: a human red blood cell looks simple under the microscope because it has lost most organelles to specialise for oxygen transport, while a liver cell is packed with mitochondria and smooth ER because it performs energy-intensive detoxification. Structure always reflects function. Teacher note: update the DQB — students should now be able to answer 'What is inside a cell?' but the question 'Why do plant cells look different?' should still be flagged as unresolved until Lesson 4.
Lesson 4 Organelle Structure & Function — The Plant Cell: How Is a Sukuma Wiki Cell Like and Unlike a Human Cell?	Students compared a side-by-side labelled diagram of an animal cell and a sukuma wiki (plant) cell. They observed and recorded three structures present in the plant cell but absent in the animal cell: chloroplasts (green, oval), a large central vacuole (occupying most of the cell's volume), and the cell wall (outermost rigid boundary). Students may also observe actual sukuma wiki cells under a microscope or projected micrograph to confirm the green chloroplasts are visible even at low magnification.	Students learned that plant cells contain three organelles NOT found in animal cells: (1) Chloroplasts — green, oval organelles containing chlorophyll that capture light energy for photosynthesis; (2) Large central vacuole — stores water, maintains turgor pressure, and stores pigments or waste products; (3) Cell wall — made of cellulose, provides rigidity and structural support. Students also reinforced that plant cells DO share the nucleus, mitochondria, ribosomes, ER, Golgi apparatus, and cell membrane with animal cells. Key vocabulary: chloroplast, chlorophyll, central vacuole, turgor pressure, cellulose, cell wall.	Students explained why sukuma wiki cells look so different from human blood cells when viewed under a microscope: the large central vacuole pushes the cytoplasm to the edges, the rigid cell wall gives the cell its box shape, and the green chloroplasts are clearly visible. All three differences are structural adaptations that allow the plant cell to perform photosynthesis and maintain structural integrity without a skeleton. Students directly answered part of the Driving Question: sukuma wiki cells look different because they have additional organelles that serve plant-specific functions. Teacher note: this is a pivotal lesson — ensure all students can list the 3 plant-only organelles and give a function for each before moving on.
Lesson 5 Are Bacteria Cells the Same as Our Cells? Prokaryotes vs.	Students observed diagrams or micrographs comparing a prokaryotic cell (e.g., Lactobacillus bacteria found in fermenting uji or mursik) with a eukaryotic	Prokaryotic cells (e.g., bacteria like Lactobacillus in fermenting uji) have no membrane-bound nucleus and no membrane-bound organelles; their DNA sits	Students explained that despite being much simpler in structure, prokaryotic cells like Lactobacillus are extraordinarily effective at their jobs — fermenting sugars in uji, fixing

Eukaryotes	cell (human cheek cell or sukuma wiki cell). They noted that the prokaryotic cell is significantly smaller, has no visible nucleus or membrane-bound organelles, has a cell wall (but not made of cellulose), and may have a capsule and flagella. Teacher should use the fermenting uji/mursik context to make bacteria feel relevant and local — these organisms are performing cellular work all around students daily.	freely in the cytoplasm in a region called the nucleoid. They do have ribosomes (smaller than eukaryotic ribosomes), a cell membrane, a cell wall (made of peptidoglycan, not cellulose), and sometimes a capsule and flagella. Eukaryotic cells (all animal, plant, and fungal cells) have a membrane-bound nucleus and specialised membrane-bound organelles. Key vocabulary: prokaryote, eukaryote, nucleoid, peptidoglycan, flagella, capsule.	nitrogen in Rift Valley soils, or causing infections. They connected this to the Driving Question: even the simplest cell's structure is perfectly matched to its function. The absence of a nucleus does not make bacteria less 'cell-like' — it makes them differently specialised. Teacher note: a common misconception is that prokaryotes have no DNA — correct this explicitly. They have DNA; it is simply not enclosed in a membrane-bound nucleus.
Lesson 6 Cell Specialisation — Why Don't All Cells Look the Same?	Students examined micrographs or diagrams of at least four specialised cells: (1) human red blood cells — biconcave, no nucleus; (2) human nerve cells (neurons) — long axons and dendrites; (3) human sperm cells — tail (flagellum), streamlined head; (4) sukuma wiki guard cells — kidney-shaped, controlling stomatal opening. Students recorded how each cell's shape deviates from the 'generic' cell model and hypothesised why before receiving explanation.	Students learned that specialised cells deviate from the generic cell model in specific, purposeful ways: red blood cells lack a nucleus to maximise haemoglobin space for oxygen transport; nerve cells have long extensions (axons) to transmit electrical signals over long distances; sperm cells have a flagellum for motility and many mitochondria to power swimming; guard cells change shape to open and close stomata, regulating gas exchange in sukuma wiki leaves. All specialisation is the result of differential gene expression during development — all these cells contain the same DNA but express different genes. Key vocabulary: specialisation, differentiation, red blood cell, neuron, axon, flagellum, guard cell, stoma.	Students explained that cell specialisation is the direct answer to the Driving Question: cells in the body (and in sukuma wiki) look different from each other because each cell type has been structurally modified to perform one specific function with maximum efficiency. No single 'generic' cell could simultaneously carry oxygen, transmit nerve signals, and photosynthesise. Specialisation requires sacrifice — red blood cells give up their nucleus; guard cells give up the typical box shape. Teacher note: link back to Lesson 3 and 4 — specialised cells amplify or remove specific organelles. This is a rich opportunity to cold-call students: 'Which organelle would you expect to find in very high numbers in a muscle cell, and why?' (Answer: mitochondria, for ATP production.)
Lesson 7 Building Our Complete Cell Model: Synthesising Evidence to Explain the Anchoring Phenomenon	Students reviewed all evidence gathered across Lessons 1–6: microscope observations, diagrams, osmosis demonstrations, prokaryote vs. eukaryote comparisons, and specialised cell micrographs. They also revisited the Driving Question Board to identify which questions had been answered and which (if any) remained. Students began constructing or finalising a 3D model (using locally available materials — cardboard, bottle tops, sukuma wiki leaves, clay, etc.) representing either an animal cell or a plant cell, with all organelles labelled and functions annotated.	Students learned how to synthesise structural and functional knowledge of all major organelles into a single coherent 3D model. They learned that plant and animal cells share a core set of organelles (nucleus, mitochondria, ribosomes, ER, Golgi, cell membrane) but differ in three key plant-specific structures (chloroplast, large central vacuole, cell wall). They reinforced that prokaryotic cells are fundamentally different in lacking membrane-bound organelles, and that specialised cells modify this blueprint for specific functions. This lesson consolidates all prior learning into a unified explanatory framework.	Students explained the full answer to the Driving Question: the cells in a human body and in sukuma wiki look different because (1) plant cells have additional organelles — chloroplasts, a large central vacuole, and a cell wall — that animal cells lack; (2) within each organism, cells specialise by amplifying, reducing, or eliminating certain organelles to maximise efficiency for a specific job; and (3) even the simplest cells (prokaryotes like bacteria in fermented uji) follow the rule that structure serves function. The 3D model serves as a physical, cumulative record of this understanding. Teacher note: during model

			presentations, require each student to point to at least two organelles and explain both structure AND function — this is the performance-level assessment of the sub-strand.
Lesson 8 Cell Biology Assessment and Reflection — Demonstrating Mastery of the Driving Question	Students engaged in a culminating assessment activity: they were presented with a novel micrograph or diagram of an unfamiliar cell type (e.g., a Kericho tea leaf mesophyll cell, a Lake Victoria fish gill cell, or an intestinal epithelial cell) and asked to identify observable structural features, infer the cell's likely function, classify it as prokaryotic or eukaryotic, and determine whether it is a plant or animal cell. This tasks students with applying — not just recalling — all learning from Lessons 1–7 to an unseen context.	Students demonstrated consolidated understanding of: (1) the defining characteristics of prokaryotic vs. eukaryotic cells; (2) the shared and distinct organelles of plant and animal cells; (3) the principle that cell structure is always matched to cell function; (4) how specialised cells modify the generic cell blueprint. Students also reflected on the full arc of the Driving Question Board, noting which questions were answered, how their thinking changed, and what new questions emerged. Key metacognitive insight: science understanding grows by revising models with new evidence — their early predictions in Lesson 1 were incomplete, and that is exactly how scientific thinking works.	Students explained novel cell examples by applying the structure-function principle developed across the sub-strand. For example: a tea leaf mesophyll cell from Kericho would be identified as a eukaryotic plant cell with many chloroplasts (high photosynthetic demand), a cell wall, and a central vacuole. A fish gill cell would be identified as an animal cell with many mitochondria (active ion transport) and a highly folded membrane (increased surface area). Students fully resolved the Driving Question: cells look different from each other — in their bodies and in sukuma wiki — because each cell's unique structure is precisely engineered for its specific function, whether that function is carrying oxygen, capturing sunlight, transmitting nerve signals, or fermenting uji. Teacher note: celebrate DQB completion as a class — physically crossing off answered questions reinforces that science is a process of building understanding over time.